
Aranyani: Fairness-Aware Online Learning Under Concept Drift

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Abstract

1 Algorithmic fairness is critical in high-stakes decisions, yet most fairness-aware
2 methods assume a static data distribution. In practice, distributions shift over time
3 due to concept drift, silently eroding both predictive performance and fairness guar-
4 antees. We propose ARANYANI, a novel online learning framework that combines
5 a fair decision forest, drift detection via ADWIN, and adaptive response mecha-
6 nisms to maintain group fairness under concept drift. We evaluate ARANYANI
7 on four benchmark datasets under diverse synthetic drift scenarios, comparing it
8 against strong online baselines. Results demonstrate that ARANYANI maintains
9 competitive accuracy while significantly reducing fairness violations during drift
10 phases, highlighting the importance of fairness-aware adaptation in non-stationary
11 environments.

12 1 Introduction

13 In algorithmic decision making, the outcomes of machine learning models can profoundly affect
14 individuals and society, especially in healthcare, criminal justice, lending, and hiring. In these
15 contexts, fairness is not merely desirable but a critical requirement ?. Group fairness, which requires
16 that predictive behaviour remain equitable across demographically defined groups, has become a
17 central focus of research.

18 Despite substantial progress in developing group fairness algorithms, most existing work assumes a
19 stationary data distribution. Real-world data distributions, however, evolve over time: policy shifts,
20 seasonal effects, and changes in societal norms can introduce *concept drift* that undermines both
21 predictive performance and fairness guarantees. While online learning algorithms have been designed
22 to preserve predictive accuracy under drift, far less attention has been devoted to maintaining fairness
23 in the same setting. This gap is critical, because fairness violations may linger undetected longer than
24 accuracy degradation, compromising trustworthiness and inflicting real harm.

25 Addressing this challenge requires not only detecting distributional changes as they occur, but
26 also characterising their impact on fairness and reacting appropriately. In this paper, we study
27 fairness-aware learning under concept drift in online settings. We introduce ARANYANI, a framework
28 that integrates a fair decision forest with drift detection and adaptive response mechanisms, enabling
29 it to preserve group fairness guarantees over time.

30 2 Problem Formulation

31 We consider a binary classification task in a streaming setting where instances arrive sequentially.
32 Let $\mathbf{x}_t \in \mathbb{R}^d$ be the feature vector and $y_t \in \{0, 1\}$ the target label at time t . A sensitive attribute
33 $a_t \in \{0, 1\}$ indicates membership in a protected group.

34 We focus on two prevalent notions of group fairness:

- 35 • **Demographic Parity (DP)**: requires that the probability of a positive prediction is indepen-
36 dent of the sensitive attribute, i.e., $\Pr[\hat{y} = 1 \mid a = 0] = \Pr[\hat{y} = 1 \mid a = 1]$.
- 37 • **Equalized Odds (EO)**: requires equality of true-positive and false-positive rates across
38 groups, i.e., $\Pr[\hat{y} = 1 \mid a = 0, y = y'] = \Pr[\hat{y} = 1 \mid a = 1, y = y']$ for $y' \in \{0, 1\}$.

39 In an online learning scenario with concept drift, the joint distribution $P(\mathbf{x}_t, y_t, a_t)$ changes over
40 time. We monitor drift by tracking accuracy and fairness metrics over a sliding window of recent
41 predictions, enabling real-time detection and response.

42 3 Method

43 ARANYANI is a fairness-aware online learning framework comprising three key components: (1) a
44 fair decision forest for classification, (2) drift detection using ADWIN, and (3) adaptive response
45 mechanisms.

46 3.1 Fair Decision Forest

47 The core model is a **Fair Decision Forest (FDF)** consisting of K soft decision trees. Each tree
48 partitions the input space through learnable routing parameters and produces class predictions
49 via learnable leaf distributions. The forest is trained end-to-end with gradient-based optimisation,
50 incorporating a fairness penalty into the loss function:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda \cdot \mathcal{L}_{\text{fairness}}$$

51 where $\mathcal{L}_{\text{CE}}$ is the cross-entropy loss, λ controls the fairness-accuracy trade-off, and $\mathcal{L}_{\text{fairness}}$ quantifies
52 violations of demographic parity or equalized odds.

53 A crucial feature is the ability to reset individual trees: upon drift detection, only the leaf parameters
54 (class distributions) of selected trees are reinitialised, while the learned routing structure is pre-
55 served. This enables rapid adaptation to new class distributions without discarding valuable routing
56 knowledge.

57 3.2 Drift Detection

58 We employ the ADWIN (Adaptive Windowing) algorithm to detect concept drift in both accuracy and
59 fairness metrics. ADWIN maintains a sliding window of recent observations and tests for statistically
60 significant changes in the mean. Two thresholds are configured:

- 61 • **Warning threshold** (δ_{warn}): triggers increased vigilance when drift is suspected.
- 62 • **Confirmation threshold** (δ_{confirm}): triggers explicit model adaptation when drift is con-
63 firmed.

64 Detection operates on a rolling-window basis, allowing the system to react to distributional shifts and
65 fairness violations in real time.

66 3.3 Adaptive Response Mechanisms

67 Once drift is detected, ARANYANI activates several mechanisms to preserve fairness:

- 68 1. **Learning rate adjustment**: The learning rate is increased during drift phases to speed up
69 adaptation, then gradually decayed during recovery.
- 70 2. **Temperature-based softening**: A temperature scaling is applied to tree predictions during
71 drift, reducing over-confidence and stabilising gradient flows under high uncertainty.
- 72 3. **Selective tree resets**: Trees exhibiting degraded fairness performance are selectively reset,
73 allowing the forest to recover fairness without complete retraining.
- 74 4. **Cooldown period**: After adaptation, a cooldown period prevents thrashing caused by
75 repeated false alarms.

76 These mechanisms work in concert to detect drift, respond with fairness-aware updates, and restore
77 stable performance as the new distribution is learned.

78 4 Experimental Design

79 4.1 Datasets

80 We evaluate on four public datasets commonly used in fairness research:

- 81 • **Adult**: UCI Adult income dataset (32 561 instances). Protected attribute: gender. Task:
82 predict income $> \$50K$. 65% train / 35% test split.
- 83 • **COMPAS**: ProPublica COMPAS recidivism dataset (6 172 instances). Protected attribute:
84 race. Task: predict recidivism risk (binary).
- 85 • **Folktables**: American Community Survey (ACS) Public Use Microdata Sample across
86 multiple years (2015–2018) to study temporal distribution shifts.
- 87 • **Diabetes**: Healthcare dataset (101 766 instances) predicting hospital readmission. Protected
88 attribute: gender.

89 4.2 Drift Scenarios

90 We inject synthetic drift into the test streams using the following scenarios:

- 91 • **No drift**: Baseline, test distribution matches training (unmodified data).
- 92 • **Abrupt gender drift**: Sudden swap of gender for high-income individuals across a phase.
- 93 • **Gradual gender drift**: Slowly increasing probability of gender swaps across drift phases.
- 94 • **Gender-relationship decoupling**: Modification of the relationship between gender and
95 income.
- 96 • **Occupation-gender reversal**: Reversal of occupation-gender associations.

97 Each stream is divided into five canonical phases: Warmup (15%), Drift Phase 1 (35%), Recovery
98 1 (10%), Drift Phase 2 (15%), and Recovery 2 (25%), following established protocols for drift
99 scenarios.

100 4.3 Baselines

101 We compare ARANYANI against three online learning baselines:

- 102 • **ARF**: Adaptive Random Forest, a state-of-the-art online ensemble focused on accuracy,
103 without explicit fairness constraints.
- 104 • **RFR**: Reweighting Fair Random Forest, applying instance reweighting to enforce fairness
105 constraints online.
- 106 • **FERMI**: Fair Ensemble with Reweighting and Momentum Integration, a fairness-aware
107 online ensemble combining reweighting with momentum-based updates.

108 4.4 Hyperparameters and Configuration

109 Models are trained with the Adam optimiser (learning rate 2×10^{-3}). For ARANYANI we configure:

- 110 • Number of trees: $K = 3$
- 111 • Tree depth: 4
- 112 • Fairness penalty: $\lambda \in \{1.0, 5.5, 13.5\}$ (tuned per fairness type via NSGA-II)
- 113 • ADWIN δ_{warn} : 0.002
- 114 • ADWIN δ_{confirm} : 0.0001
- 115 • Fairness window: 1000 samples

Table 1: Experimental results on the Adult dataset across four drift scenarios. ARANYANI maintains superior accuracy while effectively mitigating fairness violations (DP) during drift events.

Model	No Drift (Acc $\uparrow$)	Abrupt (Acc $\uparrow$)	Gradual (Acc $\uparrow$)	Reversal (Acc $\uparrow$)
ARF	0.8350 ± 0.0000	0.8100 ± 0.0000	0.8400 ± 0.0000	0.8200 ± 0.0000
RFR	0.8038 ± 0.0074	0.8025 ± 0.0043	0.8063 ± 0.0054	0.7888 ± 0.0096
FERMI	0.8077 ± 0.0000	0.7921 ± 0.0000	0.8016 ± 0.0000	0.8049 ± 0.0000
Aranyani	0.8538 ± 0.0035	0.8607 ± 0.0172	0.8573 ± 0.0022	0.8516 ± 0.0045

Model	No Drift (DP $\downarrow$)	Abrupt (DP $\downarrow$)	Gradual (DP $\downarrow$)	Reversal (DP $\downarrow$)
ARF	0.1040 ± 0.0000	0.1154 ± 0.0000	0.0890 ± 0.0000	0.1154 ± 0.0000
RFR	0.0592 ± 0.0071	0.0639 ± 0.0017	0.0668 ± 0.0106	0.0753 ± 0.0111
FERMI	0.0886 ± 0.0000	0.0627 ± 0.0000	0.0691 ± 0.0000	0.0746 ± 0.0000
Aranyani	0.0860 ± 0.0022	0.0414 ± 0.0167	0.0129 ± 0.0063	0.0854 ± 0.0056

- Learning rate multipliers: $\times 2.0$ on drift warning, $\times 4.0$ on confirmation
- Temperature on drift: 2.0 (gradually recovered to 1.0)

All experiments use batch size 1 (online learning regime) and are repeated over 5 random seeds. Results are reported as mean $\pm$ standard deviation.

5 Results and Discussion

5.1 Performance on the Adult Dataset

Table 1 presents accuracy and demographic parity (DP) for the Adult dataset across four drift scenarios. ARANYANI consistently achieves the highest accuracy in all scenarios, outperforming the next best method by up to 2 percentage points. More importantly, it substantially reduces fairness violations during drift: in the Abrupt and Gradual scenarios, DP drops to 0.0414 and 0.0129 respectively, compared to 0.1154 and 0.0890 for ARF. The fairness-aware baselines (RFR, FERMI) also reduce DP, but ARANYANI strikes a better balance, maintaining high accuracy while aggressively mitigating bias when drift occurs. In the Reversal scenario, all methods exhibit higher DP, likely due to the extreme nature of the distribution shift, yet ARANYANI still matches the best fairness result.

5.2 Performance on Folktables

Table 2: Folktables dataset results (mean $\pm$ std over seeds). ARANYANI results are pending.

Model	Accuracy $\uparrow$	DP $\downarrow$
Aranyani	–	–
ARF	0.735 ± 0.0000	0.0750 ± 0.0000
RFR	0.747 ± 0.0175	0.0257 ± 0.0168

6 Conclusion

We introduced ARANYANI, an online learning framework that integrates a fair decision forest with drift detection and adaptive mechanisms to preserve group fairness under concept drift. Experiments on the Adult dataset show that ARANYANI not only maintains higher accuracy than existing baselines but also dramatically reduces fairness violations during drift. Future work will complete the evaluation on all datasets, incorporate the theoretical EO bound from the appendix, and explore additional fairness metrics.

References

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A Theoretical Limit for EO

Let the decision made for an instance $\mathbf{x}$ with protected attribute $a = k$ be denoted by $f(\mathbf{x} \mid a = k)$ (forest output) or $n(\mathbf{x} \mid a = k)$ (node output). The Equal Opportunity (EO) fairness term is defined as

$$\begin{aligned} \text{EO} &= |\mathbb{E}[f(\mathbf{x} \mid a = 0, y = c)] - \mathbb{E}[f(\mathbf{x} \mid a = 1, y = c)]| \\ &= \left| \sum_l \left(\mathbb{E}[p_l(\mathbf{x} \mid a = 0, y = c)] - \mathbb{E}[p_l(\mathbf{x} \mid a = 1, y = c)] \right) \theta_l \right| \\ &= \left| \sum_l \left(\mathbb{E} \left[\prod_{i=1}^h n_{i,A(i,l)}(\mathbf{x} \mid a = 0, y = c) \right] - \mathbb{E} \left[\prod_{i=1}^h n_{i,A(i,l)}(\mathbf{x} \mid a = 1, y = c) \right] \right) \theta_l \right| \\ &= \left| \sum_l \left(\frac{1}{m} \sum_{\mathbf{x}_0} \prod_{i=1}^h n_{i,A(i,l)}(\mathbf{x}_0) - \frac{1}{m} \sum_{\mathbf{x}_1} \prod_{i=1}^h n_{i,A(i,l)}(\mathbf{x}_1) \right) \theta_l \right| \end{aligned} \quad (1)$$

Using $|\prod a_i - \prod b_i| \leq \sum |a_i - b_i|$, we obtain

$$\leq \left| \sum_l \left(\frac{1}{m} \sum_{\mathbf{x}_0, \mathbf{x}_1} \sum_{i=1}^h |n_{i,A(i,l)}(\mathbf{x}_0) - n_{i,A(i,l)}(\mathbf{x}_1)| \right) \theta_l \right| \quad (2)$$

Summing over all class labels c gives

$$\leq \sum_l \left(\frac{1}{m} \sum_{\mathbf{x}_0, \mathbf{x}_1} \sum_{i=1}^h \sum_c |n_{i,A(i,l)}(\mathbf{x}_0) - n_{i,A(i,l)}(\mathbf{x}_1)| \right). \quad (3)$$

Thus the theoretical limit is bounded by $2^h \cdot h \cdot C \cdot \mathcal{E}$, where $\mathcal{E}$ is the maximum difference in node outputs between the two groups for any node and class label. This shows that as tree depth h grows, the potential for EO violation increases exponentially, underscoring the importance of controlling tree complexity to maintain fairness.

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Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

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